

Power series — 6.1

Definition: A series of the form

$$\sum_{n=0}^{\infty} c_n x^n = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + \dots$$

is called a power series. Specifically, this is a power series centered at $a=0$.

We've already studied a particular power series, namely geometric series

$$\sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \dots, \quad \text{and on } (-1, 1),$$

this is equal to $\frac{1}{1-x}$.

More generally, a power series centered at a
is a series of the form

$$\sum_{n=0}^{\infty} c_n (x-a)^n.$$

Examples: $\sum_{n=0}^{\infty} \frac{x^n}{n!}$ is a power series

centered at 0.

$\sum_{n=0}^{\infty} \frac{(x-2)^n}{(n+1)3^n}$ is a power series

centered at 2.